
Overview of FPD and Driving Method

2003. 7. 18

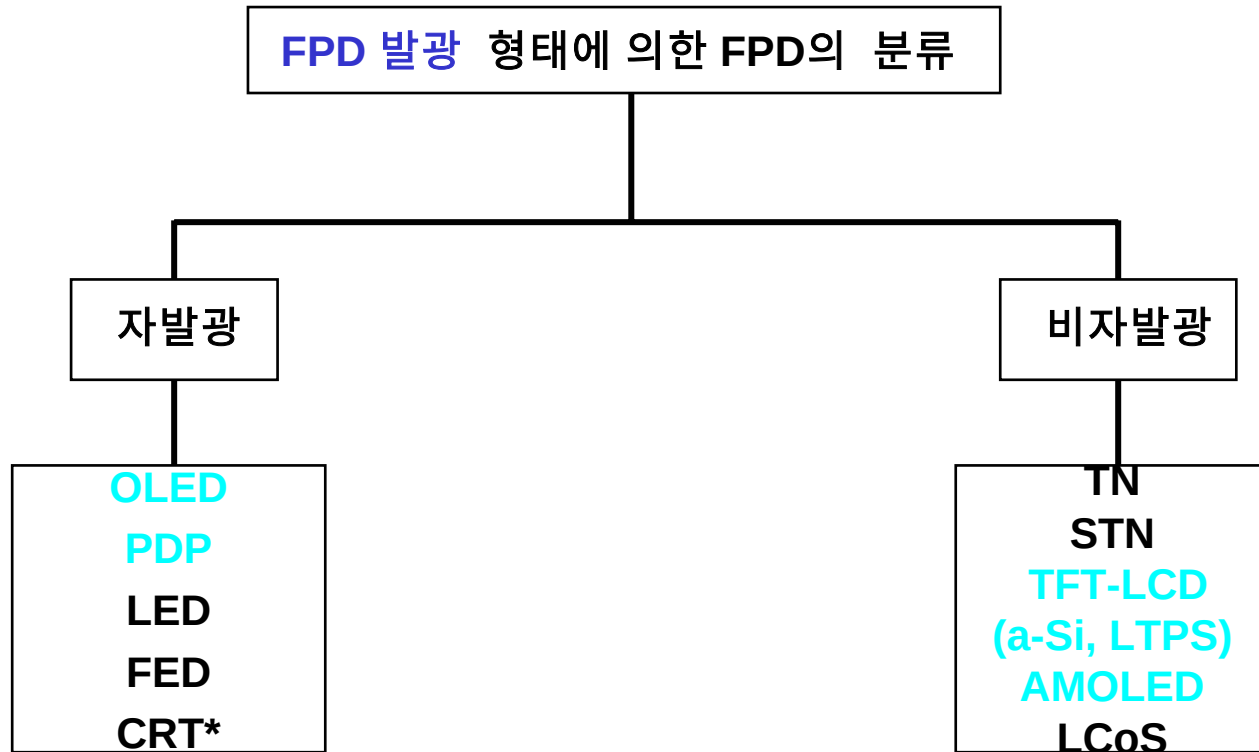
이 형 수

DNI Group, SIC Division, LG Electronics

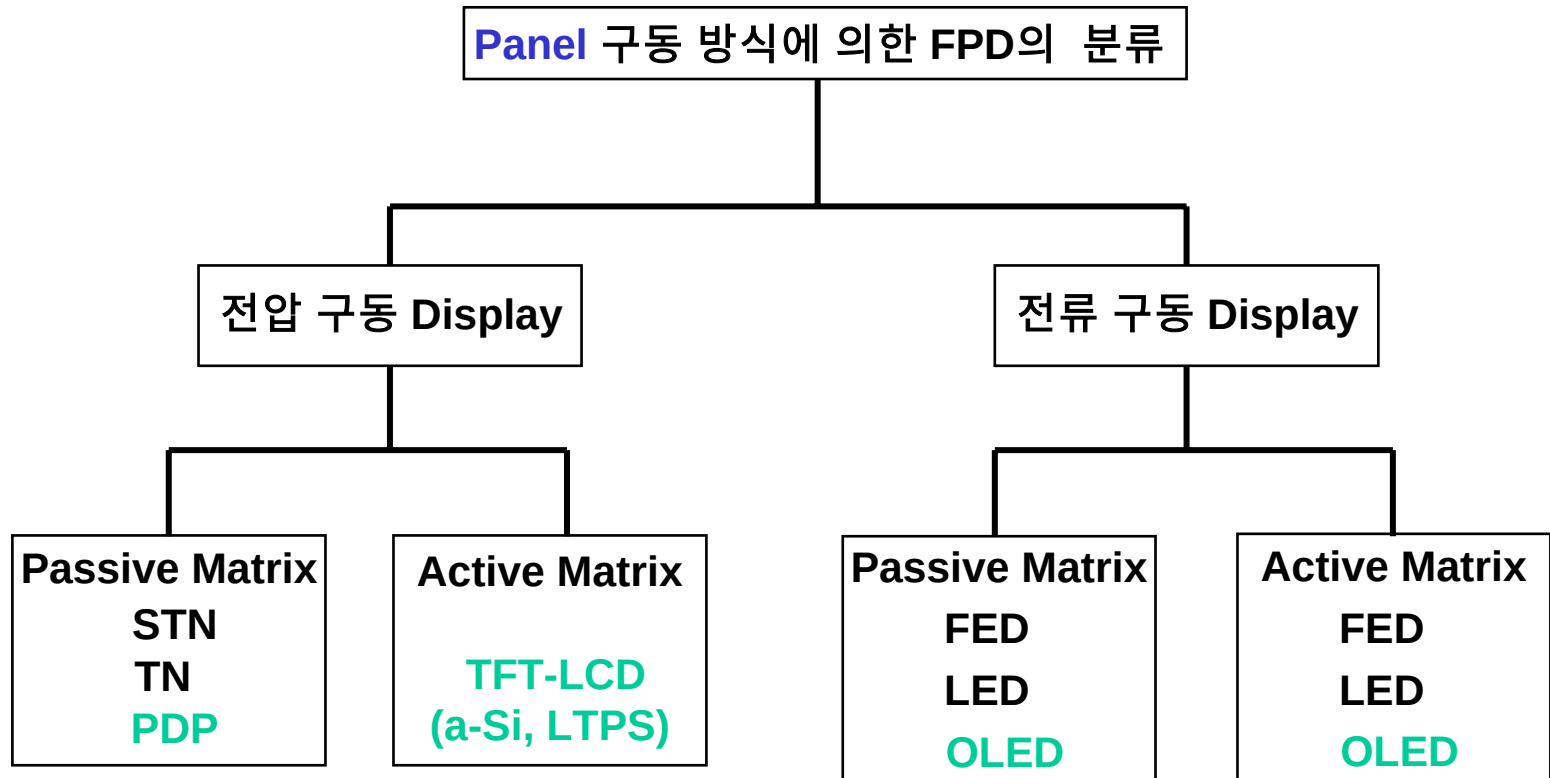
발표 순서

1. Overview of FPD
2. TFT-LCD
3. OLED
4. PDP
5. Conclusion

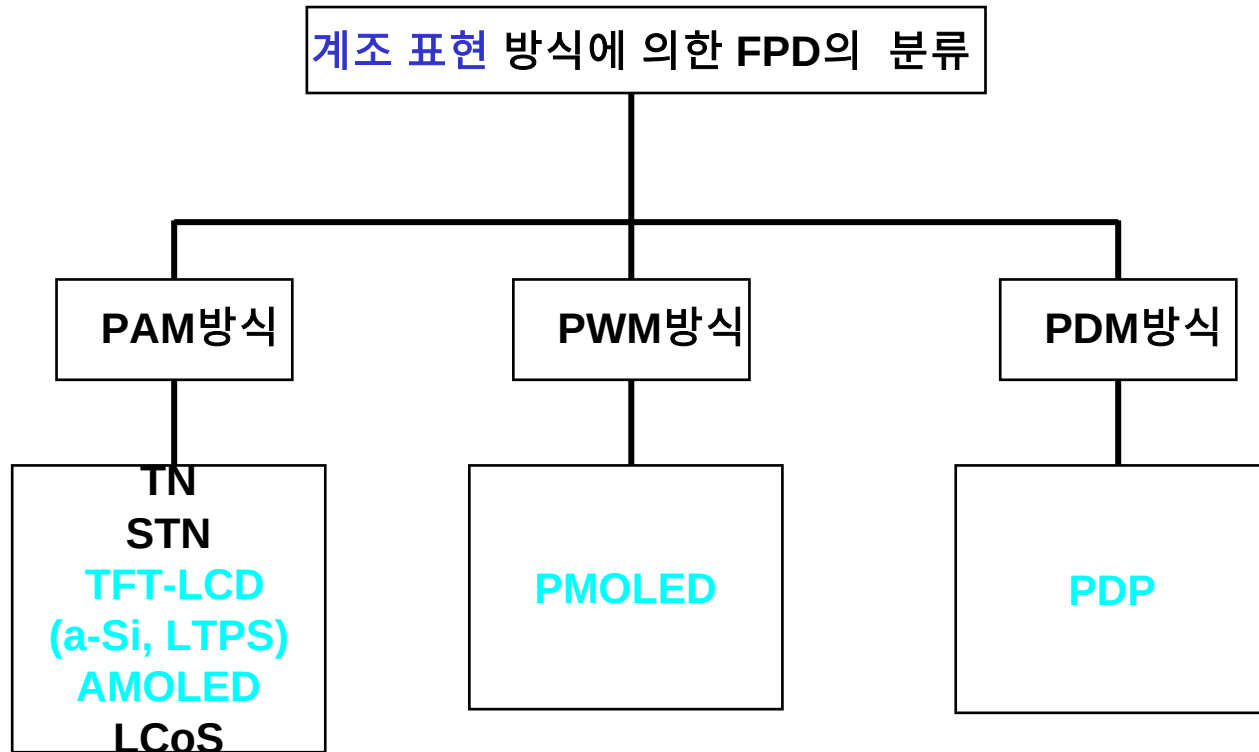
1. Overview of Flat Panel Display



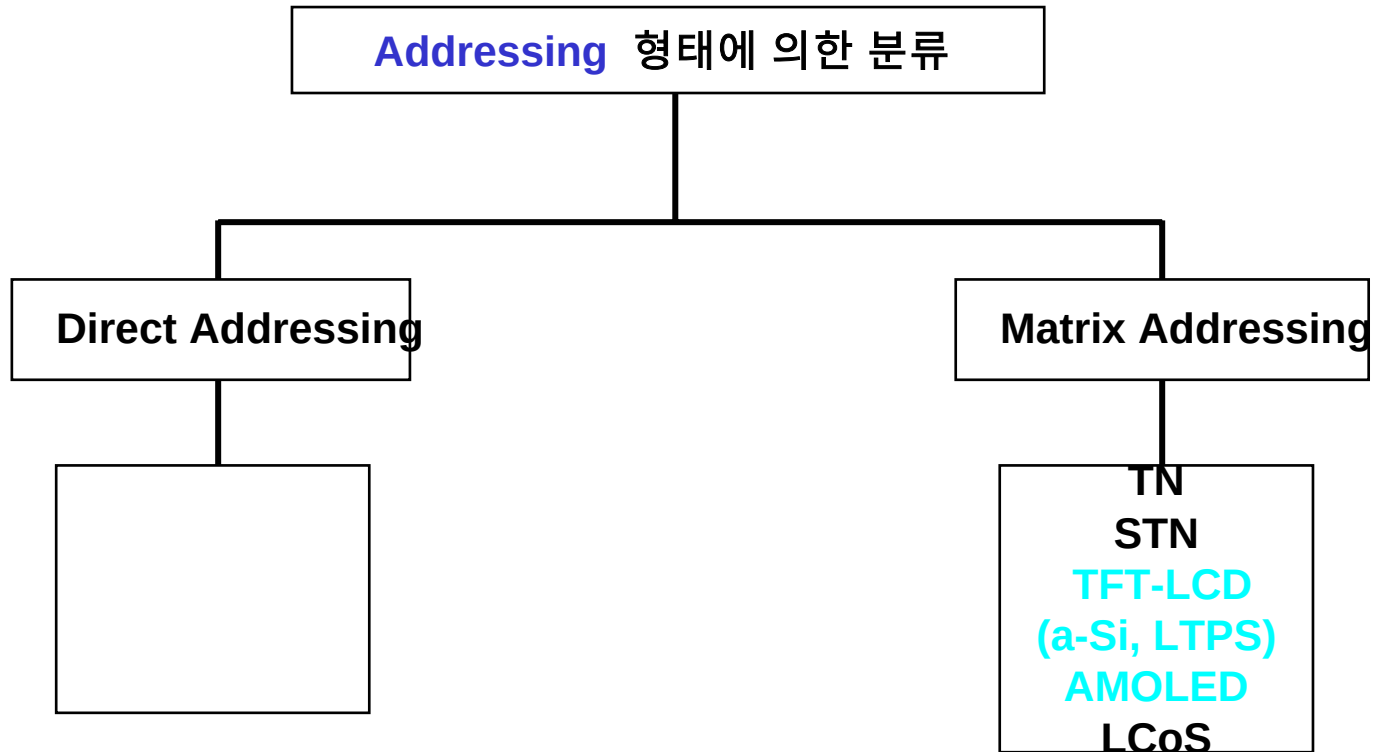
1. Overview of Flat Panel Display



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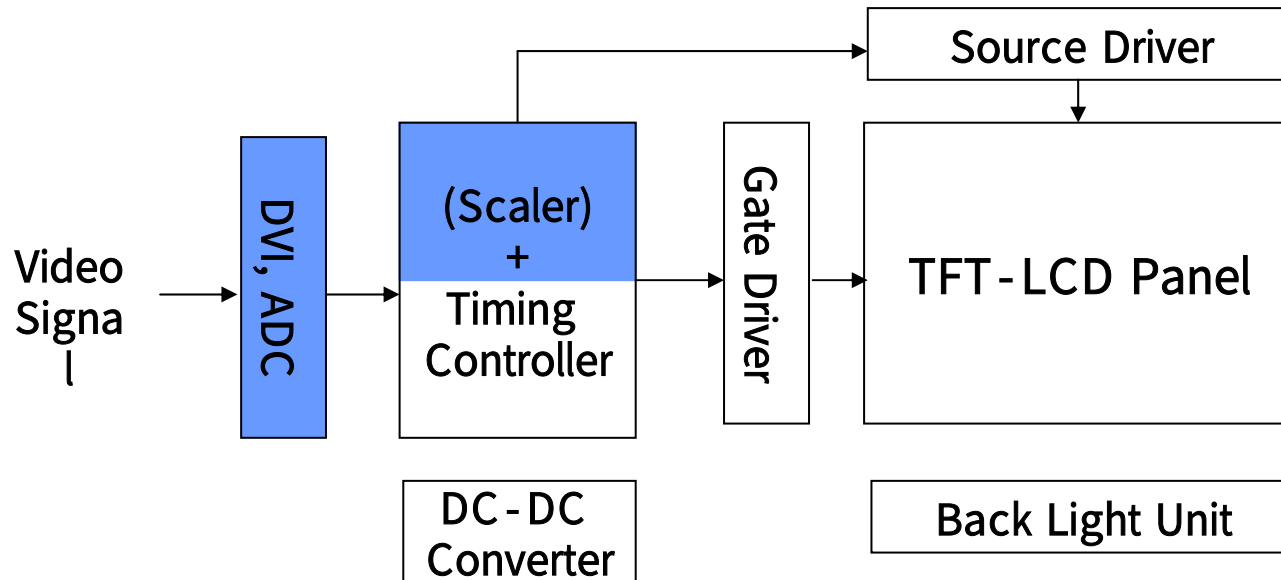


1. Overview of Flat Panel Display



2. TFT - LCD Display

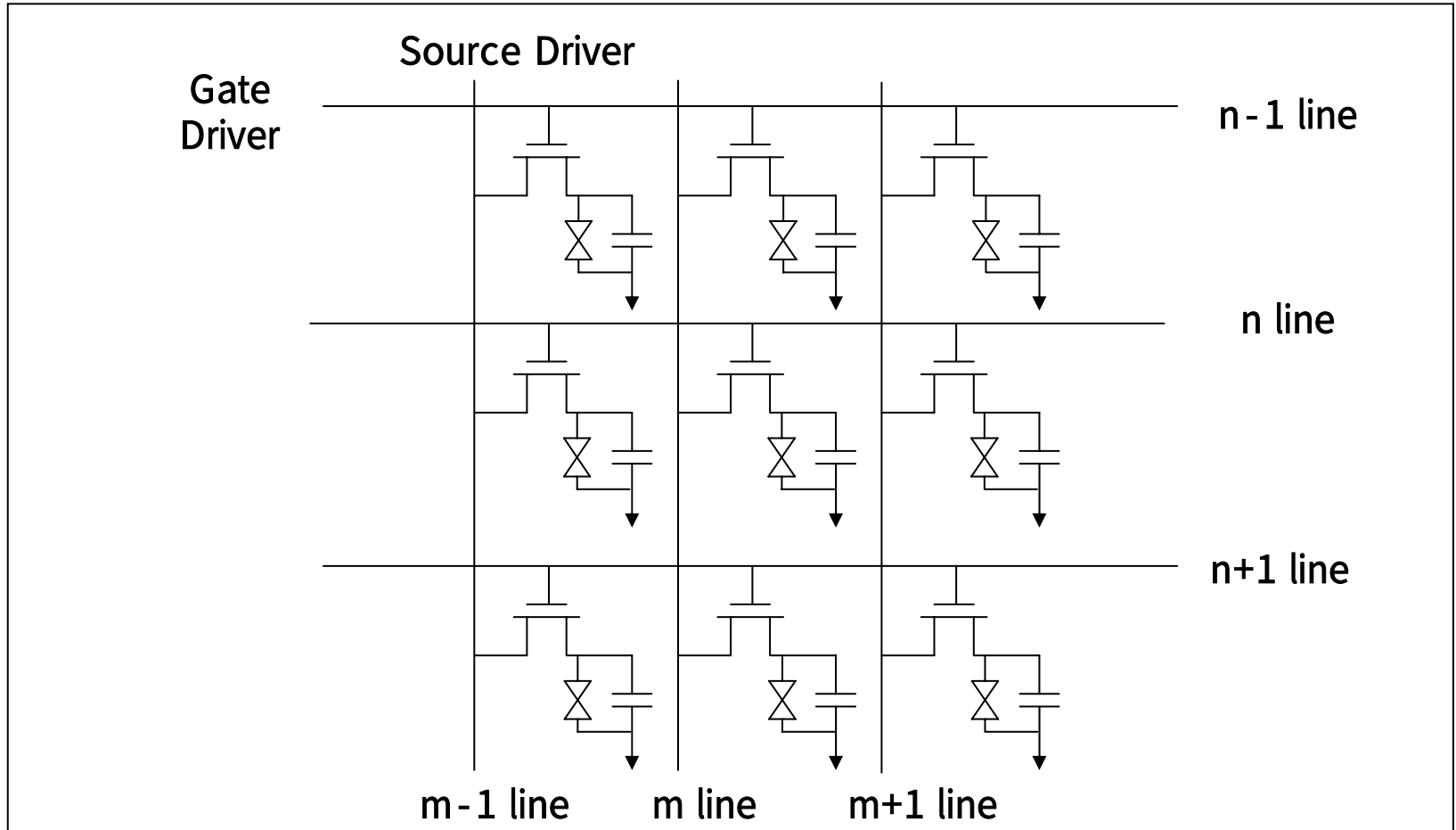
TFT-LCD Display의 구성



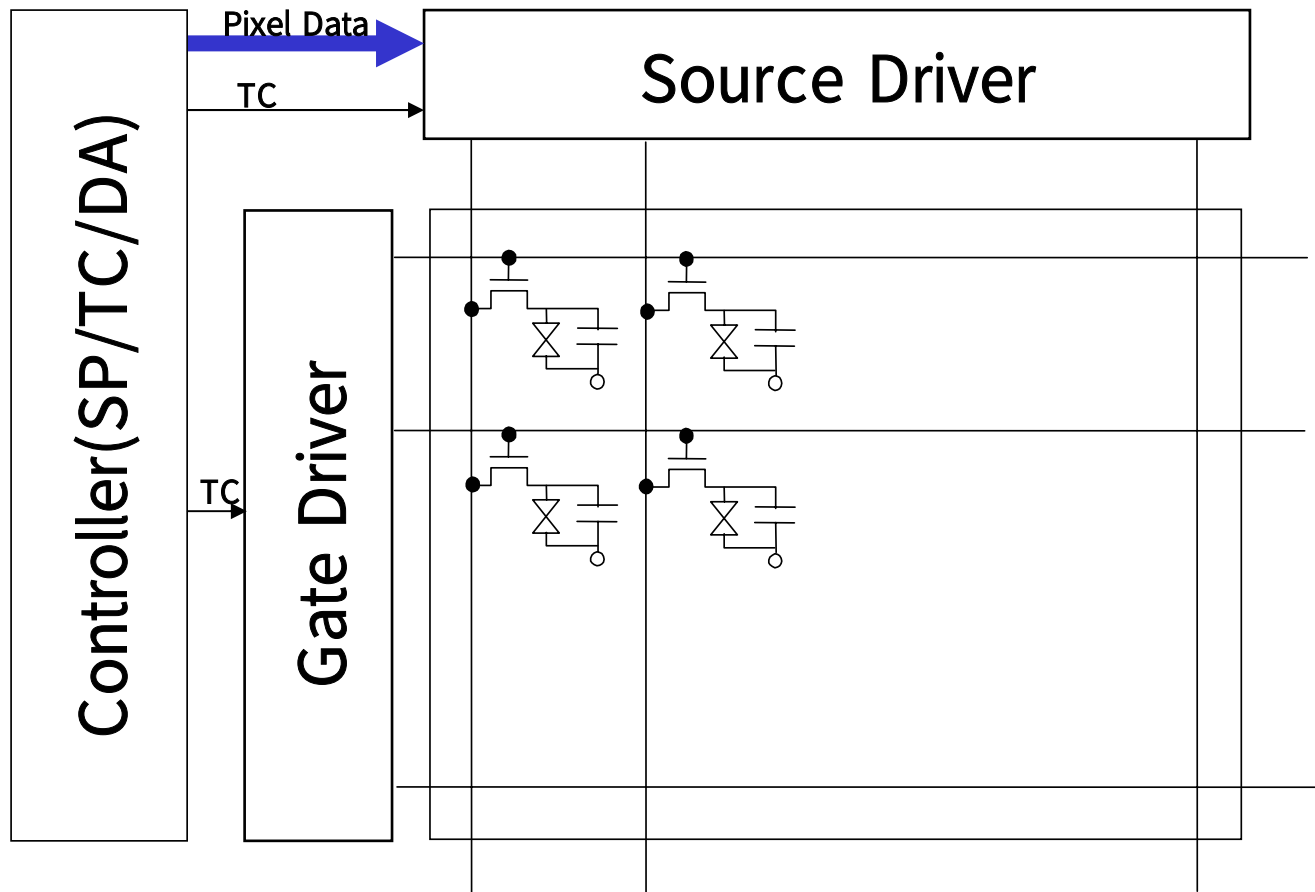
■ 중대형 Display에 필요한 기능 , 소형 Display에는 필요 없음

2. TFT - LCD Display

TFT-LCD Pixel의 구조

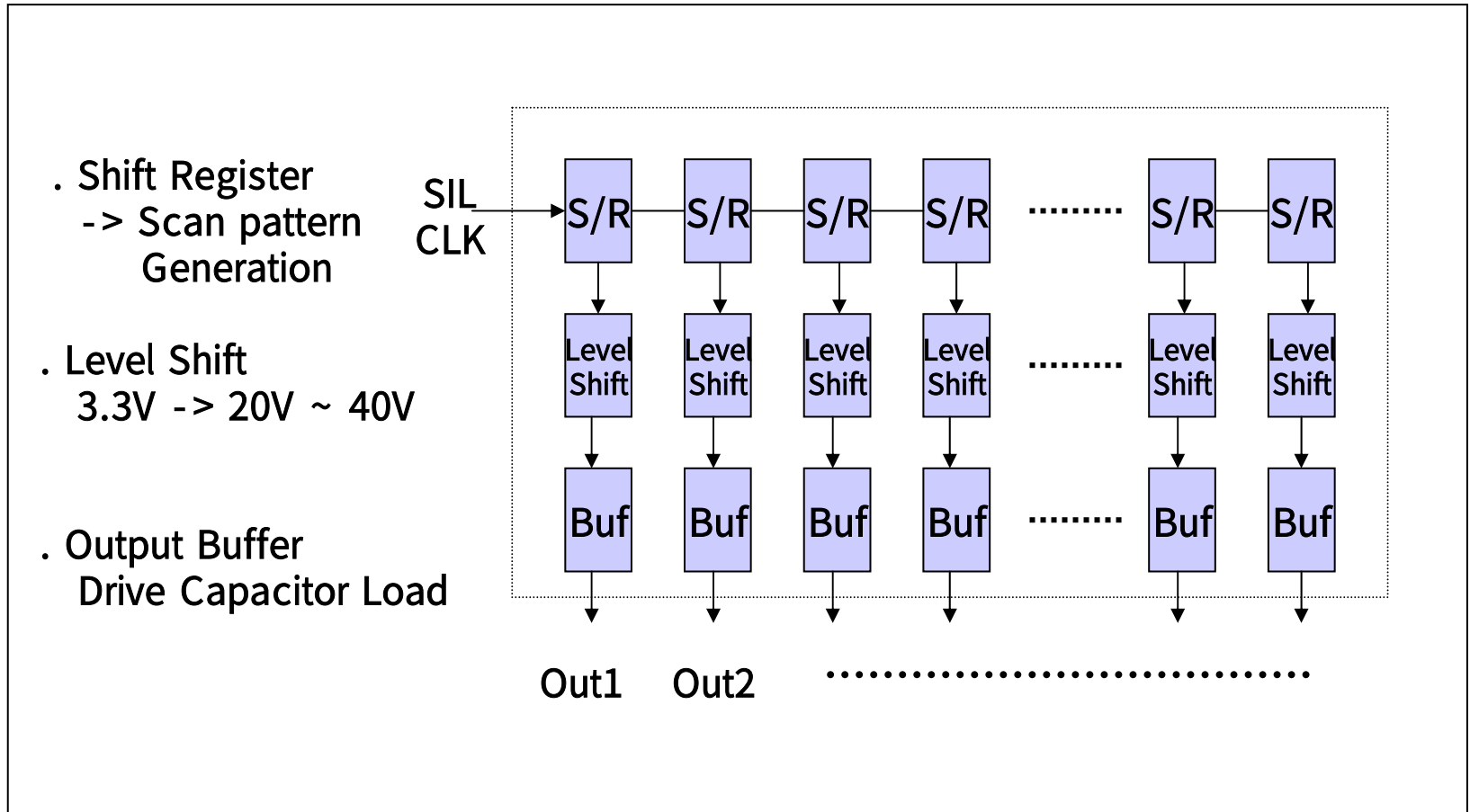


2. TFT - LCD Display

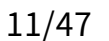


2. TFT - LCD Display

Block Diagram of Gate Driver

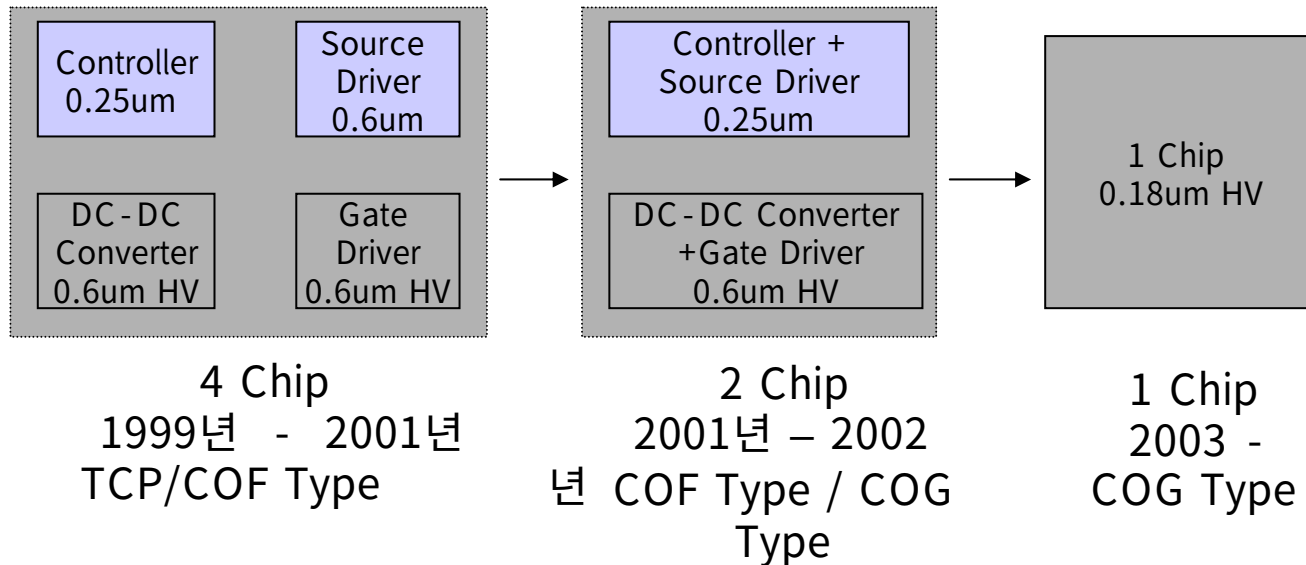


Block Diagram of Source Driver



2. TFT - LCD Display

소형 TFT LCD 구동 Chip 구성의 변화



- TFT LCD 구동 Chip의 제조 공정이 0.6um, 0.35um에서 0.18um로 옮겨가고 있음.
- 4 Chip 구성 -> 2 Chip 구성 -> 1 Chip으로 구성이 변화됨
- Package 는 COG 형태가 주류를 이루고 있음

2. TFT - LCD Display

a - Si TFT LCD와 Poly - Si TFT LCD 의 구성 비교



a-Si TFT-LCD Module

Poly-Si TFT-LCD Module

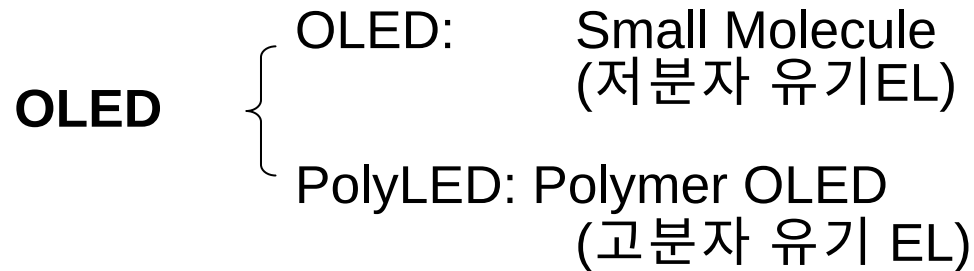
Gate Driver / Demux are Integrated
Simple Source Driver
Low Panel Yield, High Cost
Suitable for High Resolution Mobile Display

2. TFT - LCD Display

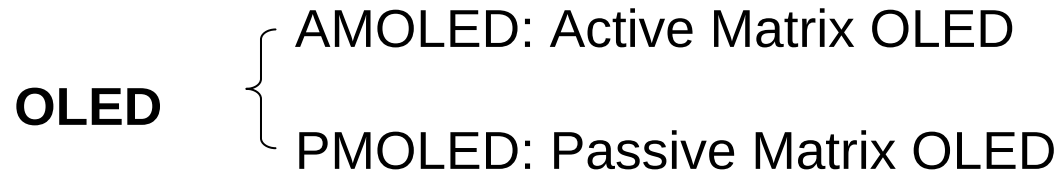
Technical Issue for TFT-LCD Driver

- Area
 - Compact I/O Cell (Small Pin Pitch, High ESD Voltage)
 - Compact SRAM (Timing Controller is integrated)
 - Compact DAC Driver
- Uniformity
 - Offset Cancellation
 - Small Gate Voltage Feed Through & Compensation
- Power
 - Small DC Current & Power Save Structure

Material 에 의한 분류



구동 방식 에 의한 분류



*OLED : Organic Light Emitting Diode

3. OLED Display

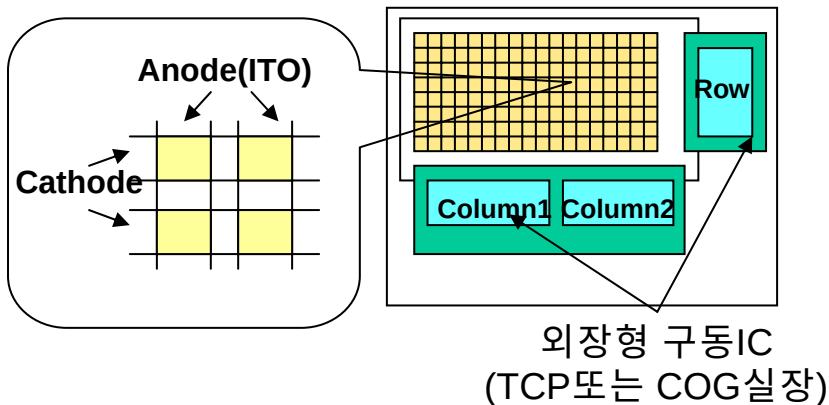
OLED DEVICE 의 특

- 응답속도가 빠르다 (a few nano Sec)
- Contrast Ratio가 좋음
- 자발광이어서 Viewing Angle이 크다
- Color Filters 없이 Color 구현 가능
- 저전압구동이 가능함
- 가볍고 얇아 Mobile Display에 적합함
- 발광량과 전류량은 거의 비례한다

3. OLED Display

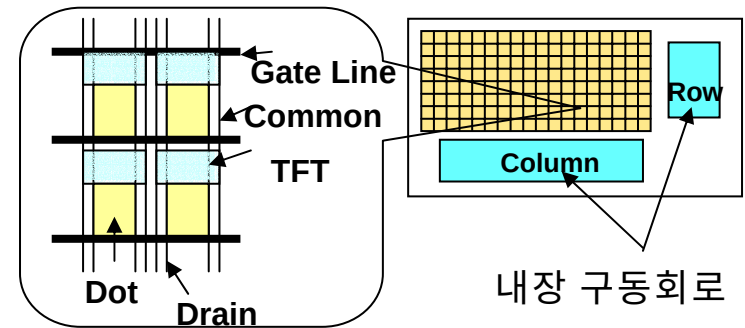
AM 방식과 PM 방식의 비교

Passive Matrix



- 화소는 단순 Matrix 구성 (양극과 음극의 단순교차구조)
- Row Line 선택시에 점등 -> Duty 구동
- 주변회로는 외장
- 특징: 단순공정

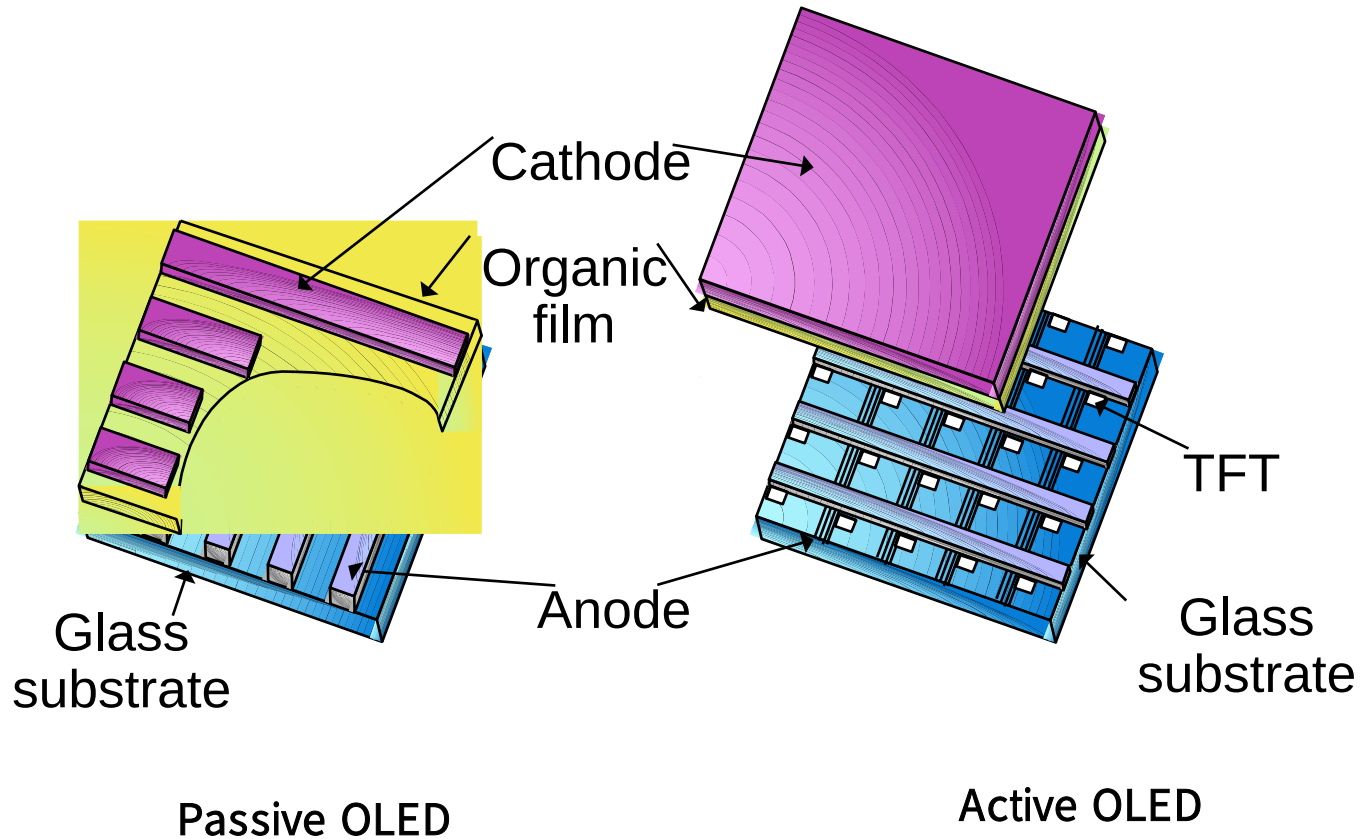
Active Matrix



- 각 화소에 Switching용 TFT를 배치
- Gate Drive와 Demux 회로 내장
- 특징: PM 방식에 비해 대화면 용이,
저소비전력

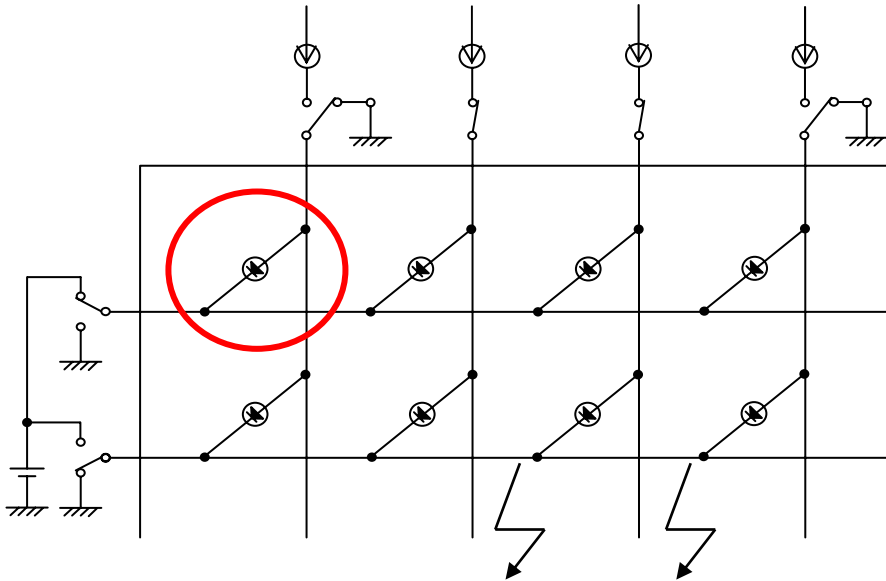
3. OLED Display

Active 와 Passive 의 기본 구조 비교

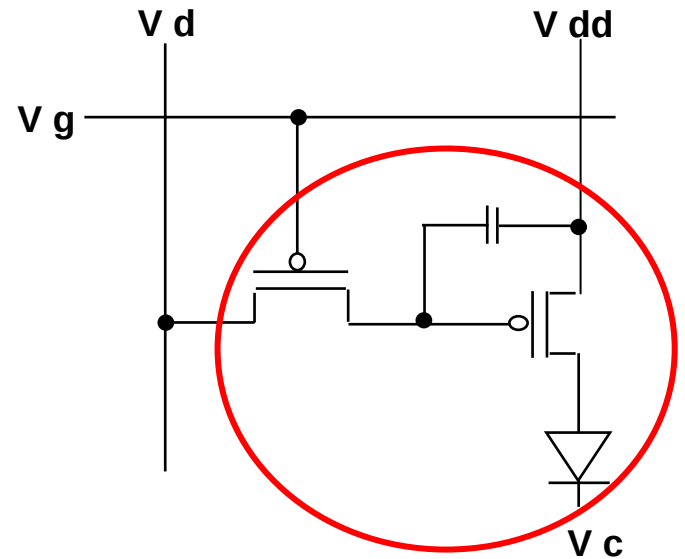


3. OLED Display

Active 와 Passive 의 구동 원리 비교



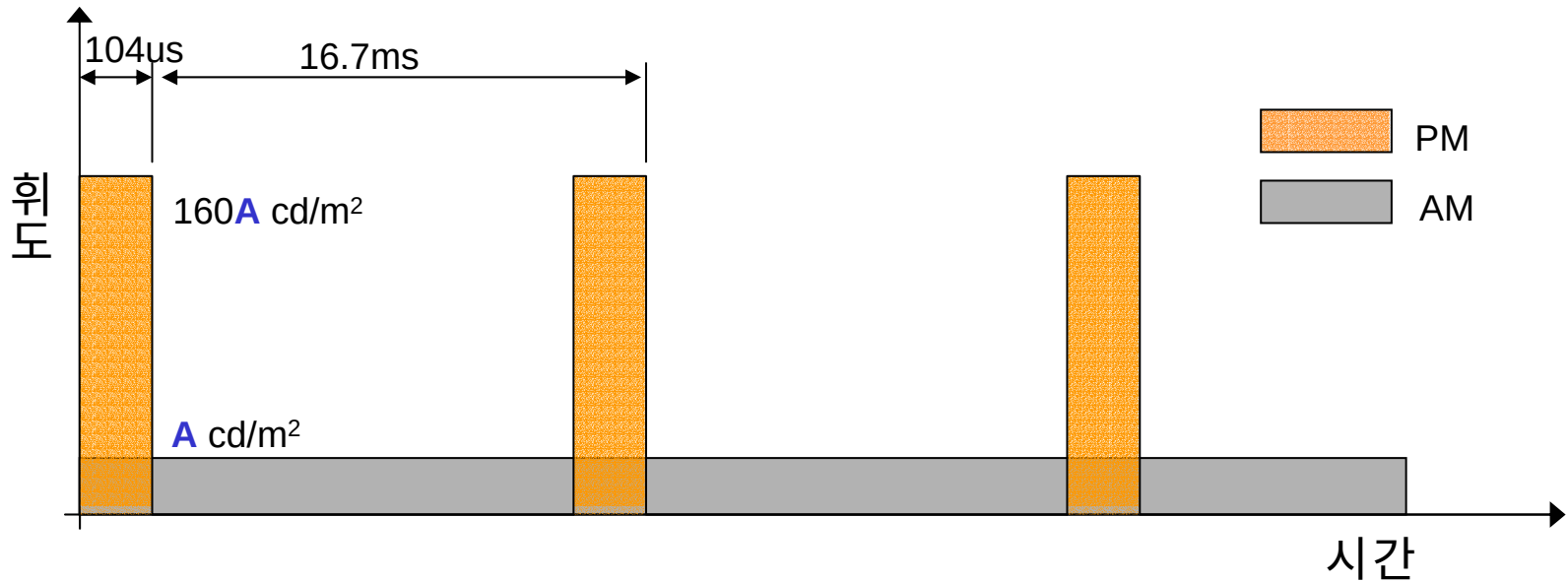
Passive OLED



Active OLED

3. OLED Display

AM OLED와 PM OLED의 발광 방식 비교



1 Frame Time : 16.7 ms(60Hz)

1 line scan time : 104 us(160 line in 1 frame)

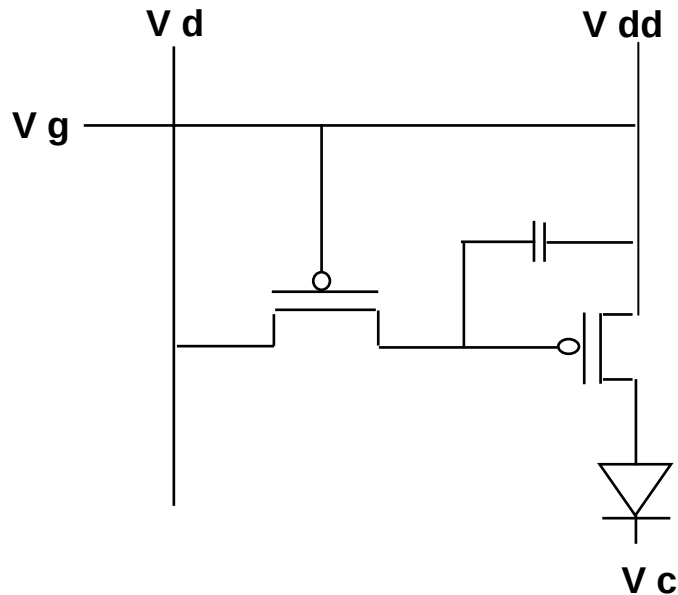
Duty Driving에 필요한 휘도 = $A \times 16.7\text{ms} / 104\text{us}$ 160A cd/m²

3. OLED Display

PM OLED의 구동 방식

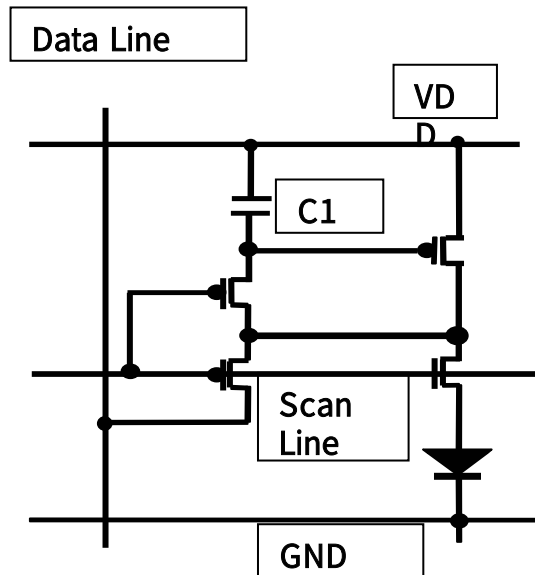
- PWM (Pulse – Width Modulation) 방식
 - Constant Current 구동
 - Variation of Time
 - > 작은 Gray Scale 를 정확하게 표시하는 것이 어려움
 - . Rising Time / Falling Time 의 Unbalance 의 영향
 - . Threshold 근처에서 OLED 발광량을 control 하기 어려움
- PAM (Pulse – Amplitude Modulation) 방식
 - Constant Time 구동
 - Variation of Current
 - > Channel Variation Issue, Layout Complexity

One capacitor and two p-type poly-Si TFT's



- 구동 방식 : 전압 구동 방식
- 특징
 - 구조가 간단함.
 - Poly TFT의 V_{th} Variation 영향에 따른 Uniformity 문제

One capacitor and 4 poly-Si TFT's



• 구동 방식 : 전류 구동 방식

• 특징

- V_{th} Variation과 무관
-> Current Programming
- Current 제어가 어려움
-> Precharge Time, Leakage Current
- Large Layout Area (개구율이 낮음)

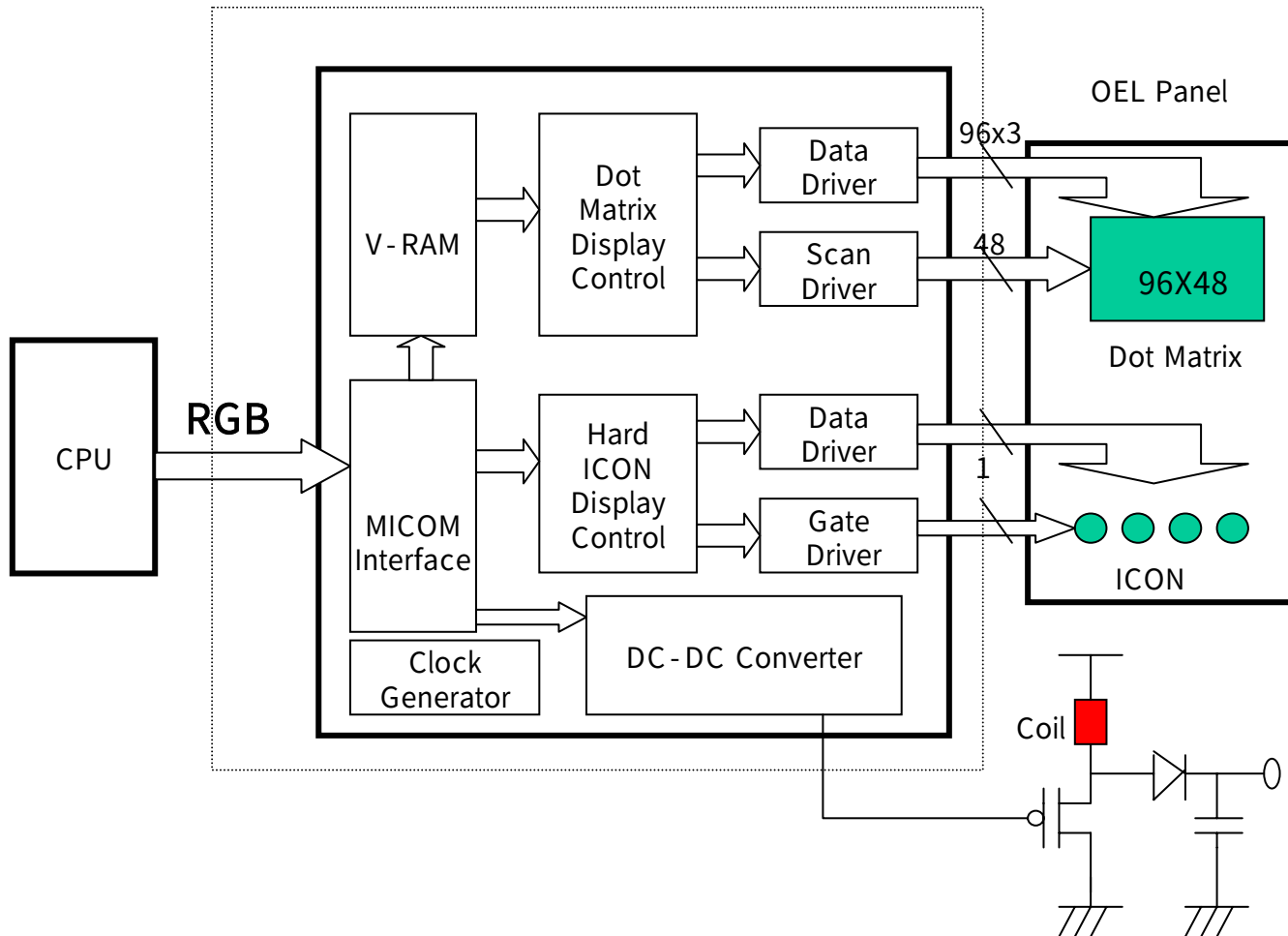
3. OLED Display

PMOLED와 AMOLED의 비교

	PMOLED	AMOLED
구동전류의 제어방법	PWM / PAM 방식	내장TR의 Vgs 전압
Pixel 당 필요한 전류치	30uA ~ 200uA	1uA 미만
Pixel 의 발광기간	Frame기간 × 1/주사선수	전Frame기간
소모 전력	300mW~400mW (100cd,2"급)	150mW(100cd,2"급)
제작 방법	간단	복잡
Cost	Panel Cost가 낮고 Driver IC Cost가 높음	Panel Cost가 높고 Driver IC cost 낮음
적용 분야	소형 Display (3" 이하)	소형 및 중형 Display

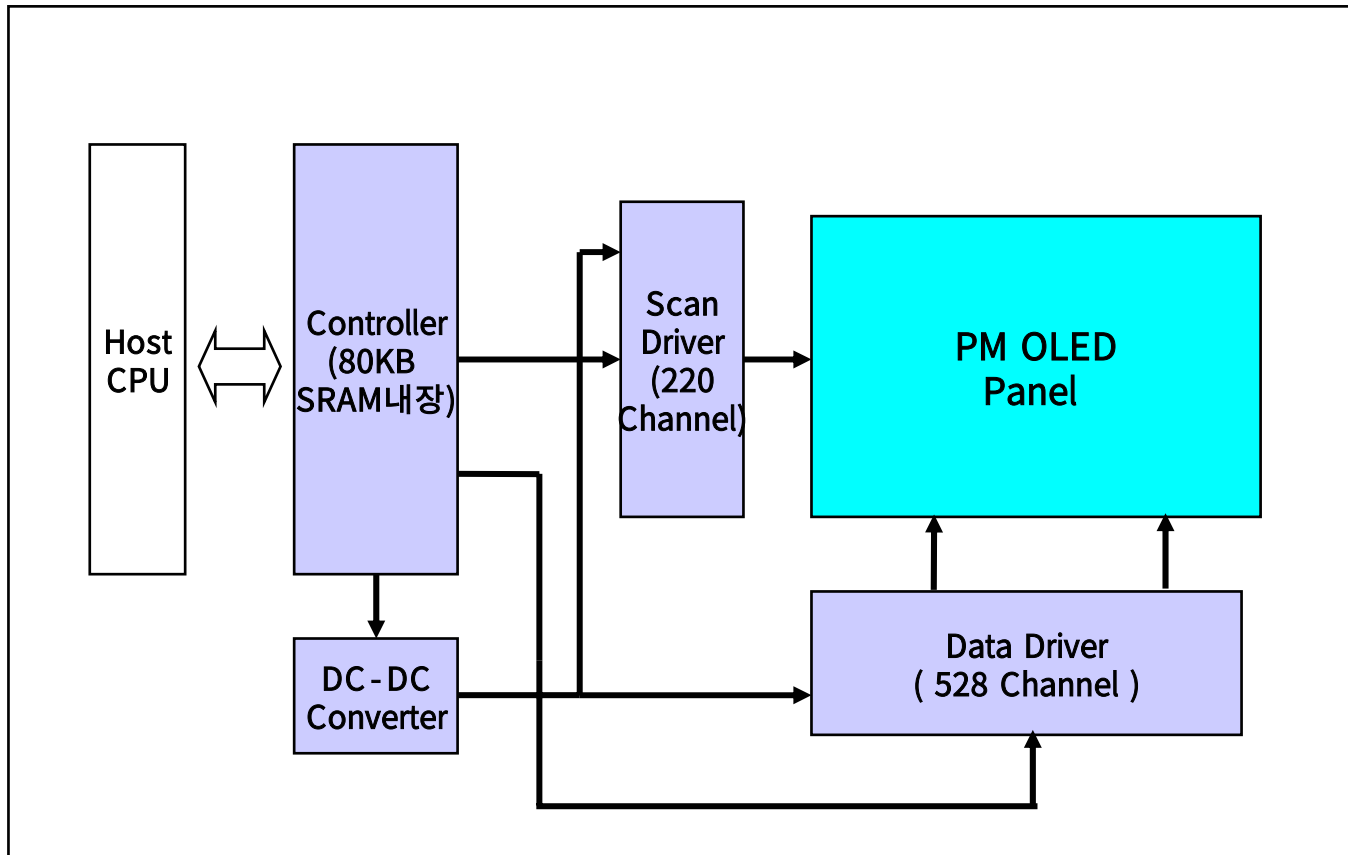
3. OLED Display

PM 방식의 OLED 구동 System Block Diagram (Single Chip)



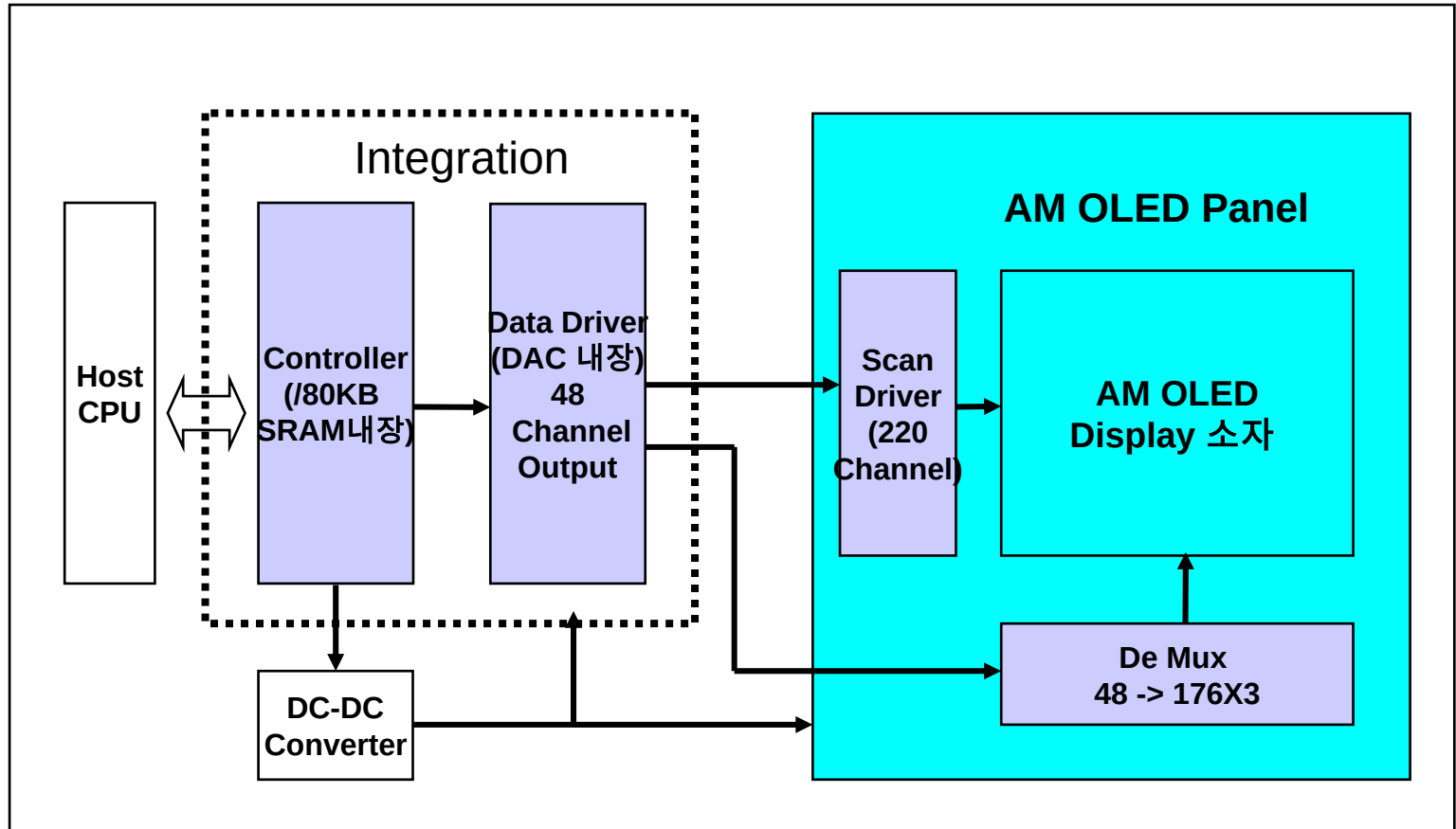
3. OLED Display

PM OLED Display Module(176X220) 구성도



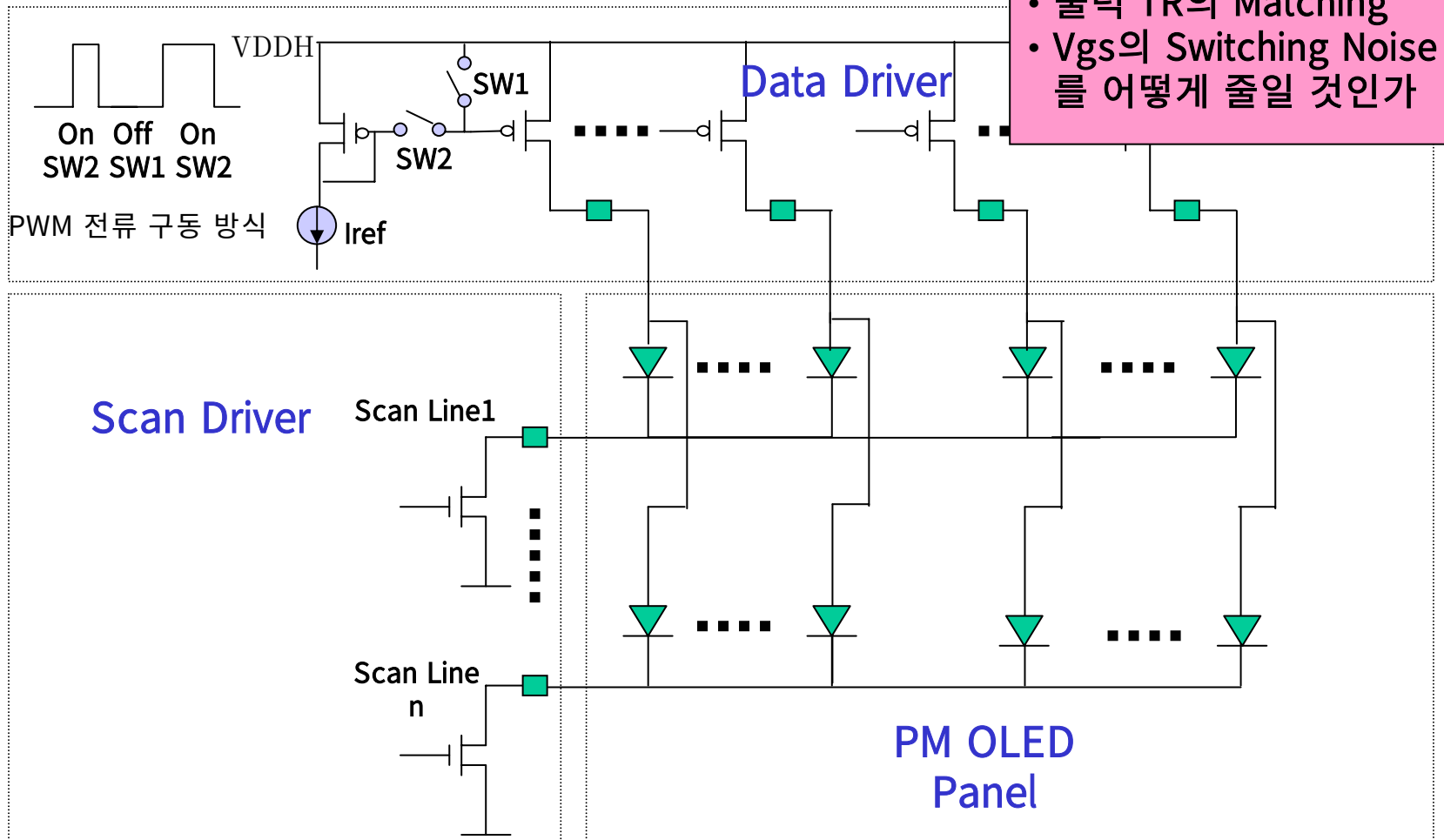
3. OLED Display

AM OLED Display Module 구성도



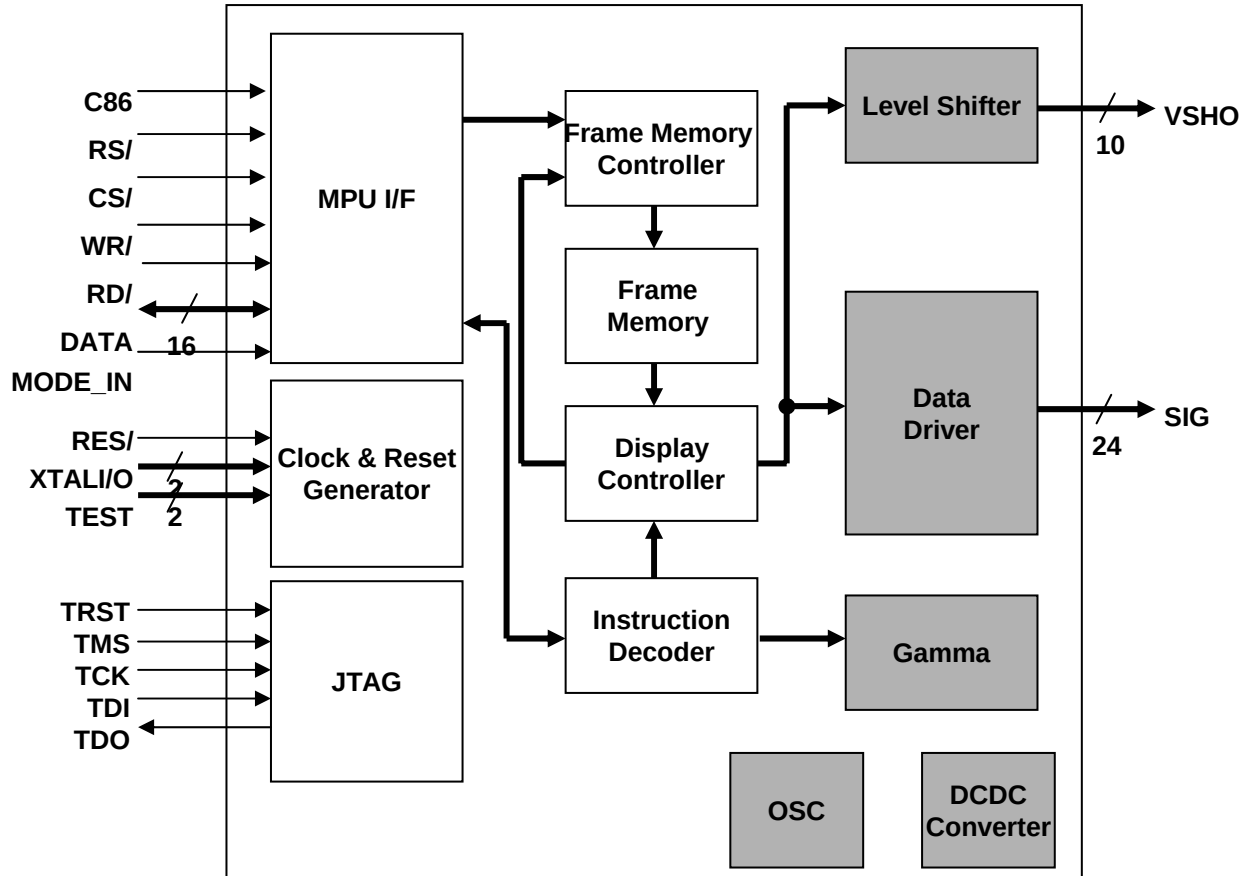
3. OLED Display

PM 방식의 OLED 구동 회로

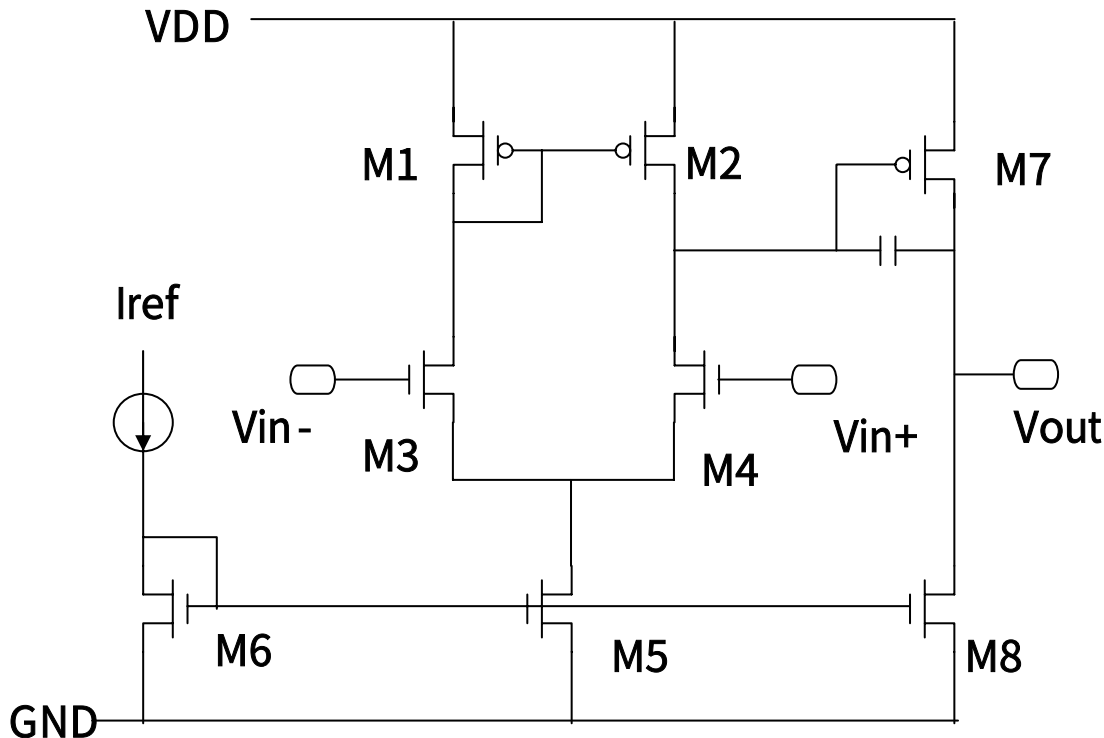


3. OLED Display

AM 방식의 OLED 구동 System Block Diagram (Single Chip)



OUTPUT Buffer Example 1.



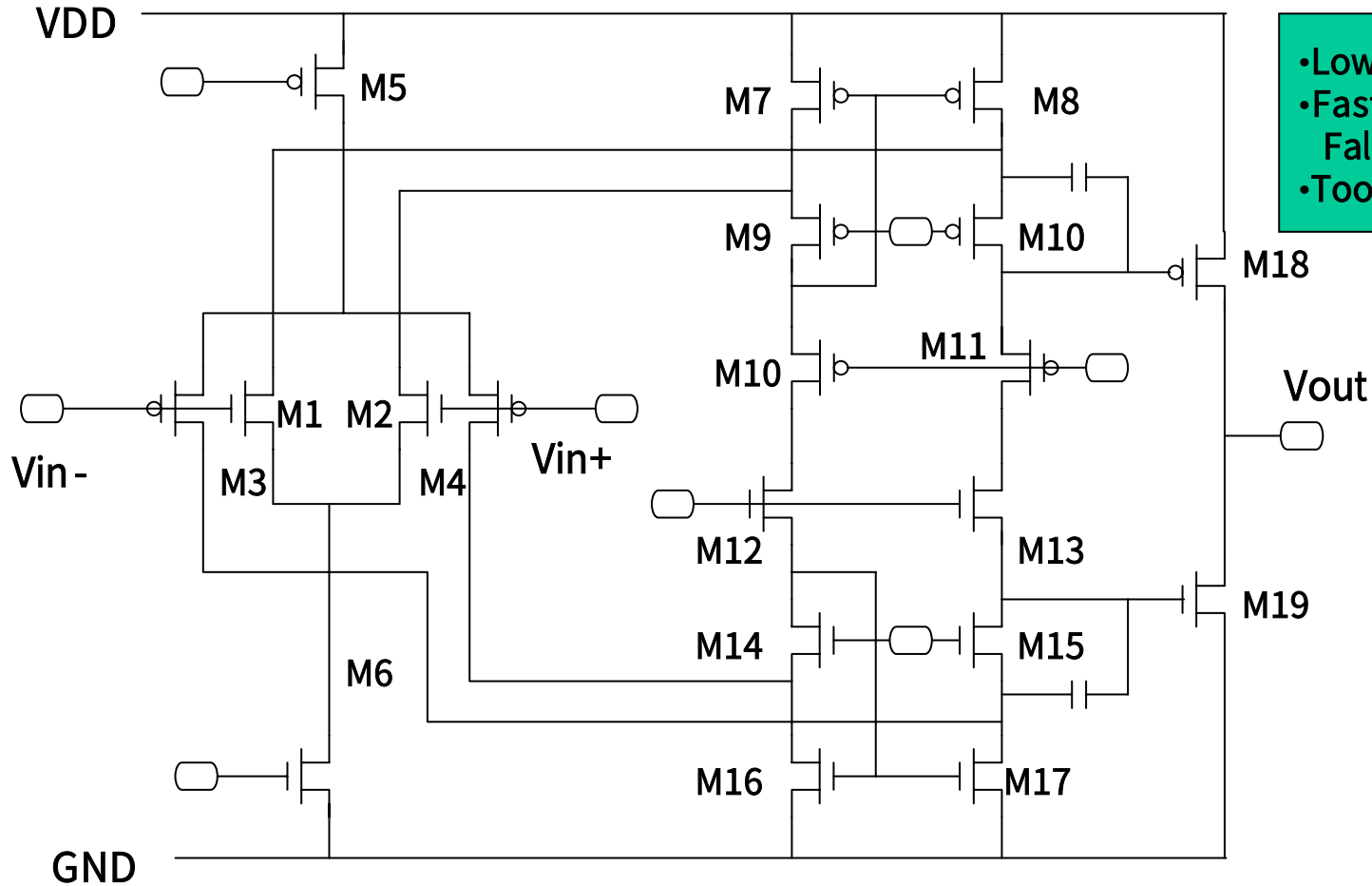
Problems

- Slow Falling Time
- Large Bias Current
- The size of M8 must be large

3. OLED Display

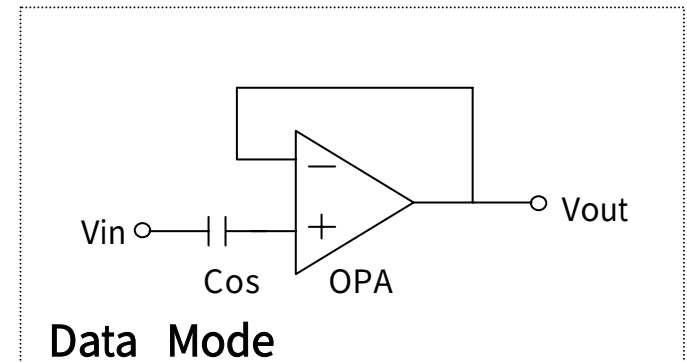
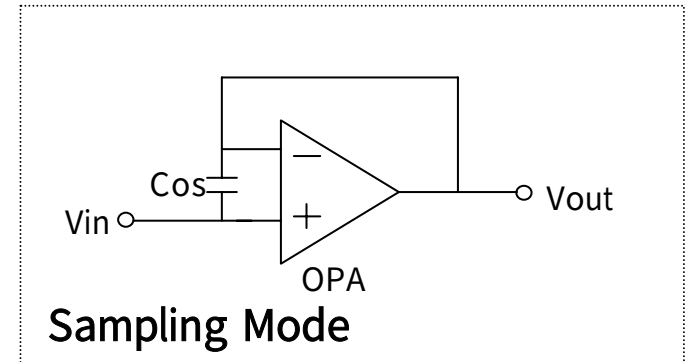
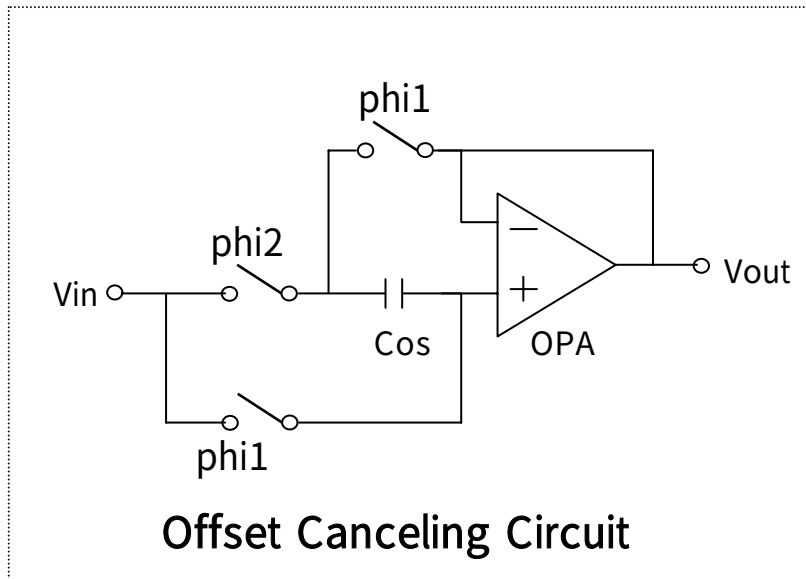
Driving Circuit

OUTPUT Buffer Example 2



- Low Bias Current
- Fast Rising & Falling Time
- Too Many Bias

Offset Canceling to Reduce Channel Difference



3. OLED Display

Technical Issue for OLED Display

- 수명 개선
 - 상품화를 위하여 수명을 수천시간에서 10,000시간 이상으로 개선
- 구동 방식의 Optimize
 - 작은 전류 구동시 Current Source의 Leakage와 Variation Control
 - Pre_Charge Issue
 - Uniformity
- 발광효율 개선
 - 물질 개선 (image sticking)

3. OLED Display



LG전자 OLED Display Phone Sample

4. PDP Driver

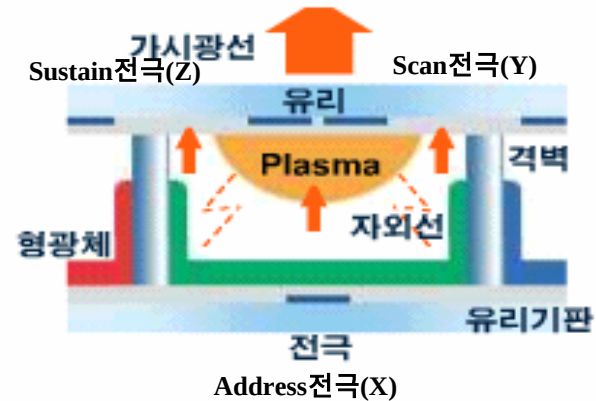
PDP 동작 원리

단계1) X전극과 Y전극에 펄스형태의 전위차 형성방전

단계2) Y전극 Z전극 방전유도

단계3) 방전과정에서 생성된 진공자외선에 의하여 R.G.B의 형광체의 발광

단계4) 빨강색(R),녹색(G),파란색(B) 광조합의 시청



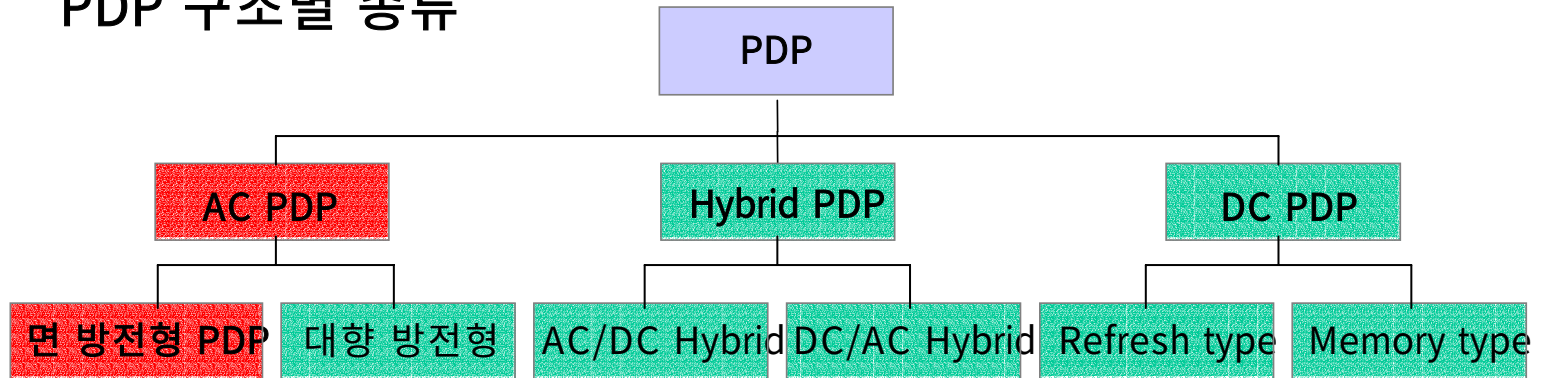
기존의 일반 브라운관(CRT)TV는 브라운관 맨 뒤에 있는 전자총이 주사(scan)하여 화면을 나타내는 원리이지만, PDP(Plasma Display Panel)는 두 장의 얇은 유리기판 사이에 혼합 가스를 주입하고 전압을 가하여 네온광이 발광되는 현상을 이용한 표시(디스플레이)

장치를 말합니다.

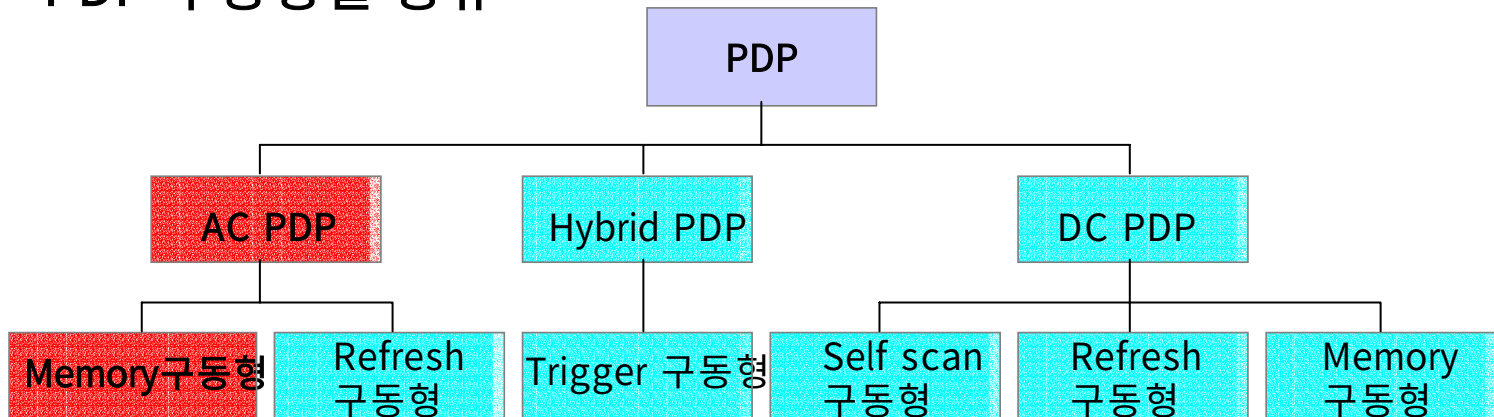
쉽게는 아주 미세한 빨강색,녹색,파란색(赤·綠·靑)의 형광등을 무수히 많이 배열해 각각의 형광등을 빠르게 점등 또는 소등시킴으로써 화면을 구현한다고 할 수 있습니다.

4. PDP Driver

PDP 구조별 종류

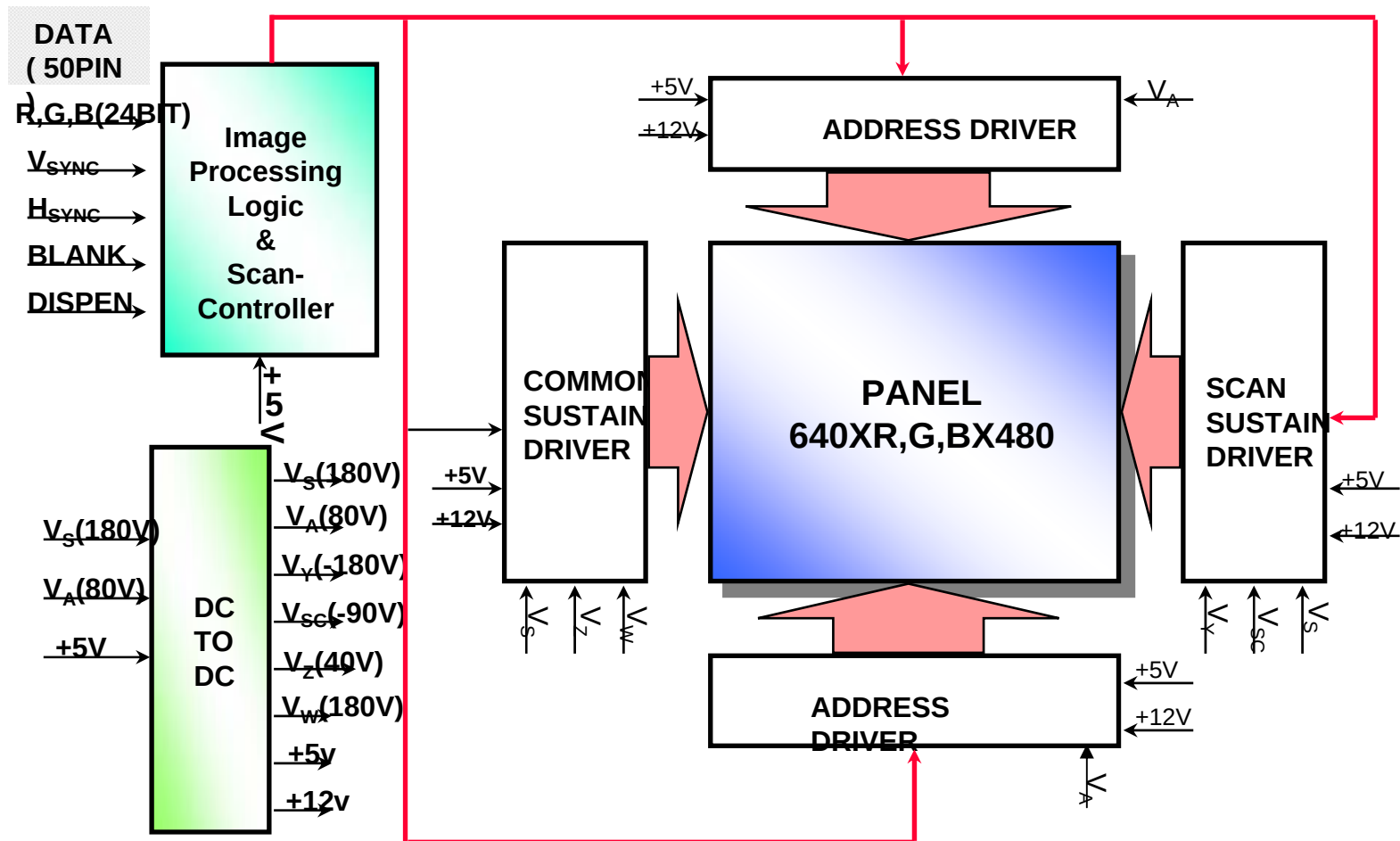


PDP 구동형별 종류



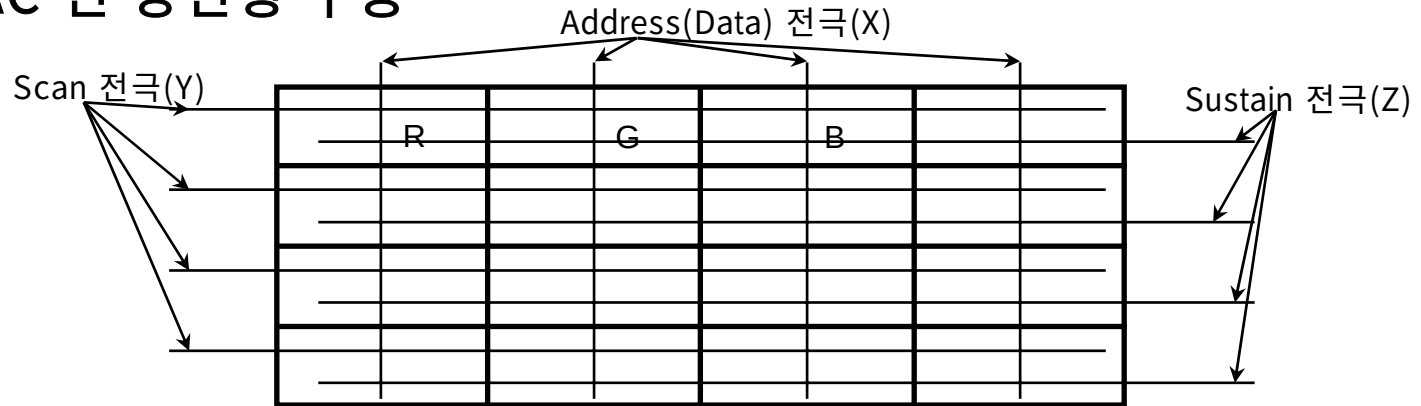
4. PDP Driver

PDP Module 구성 예(40 Inch)

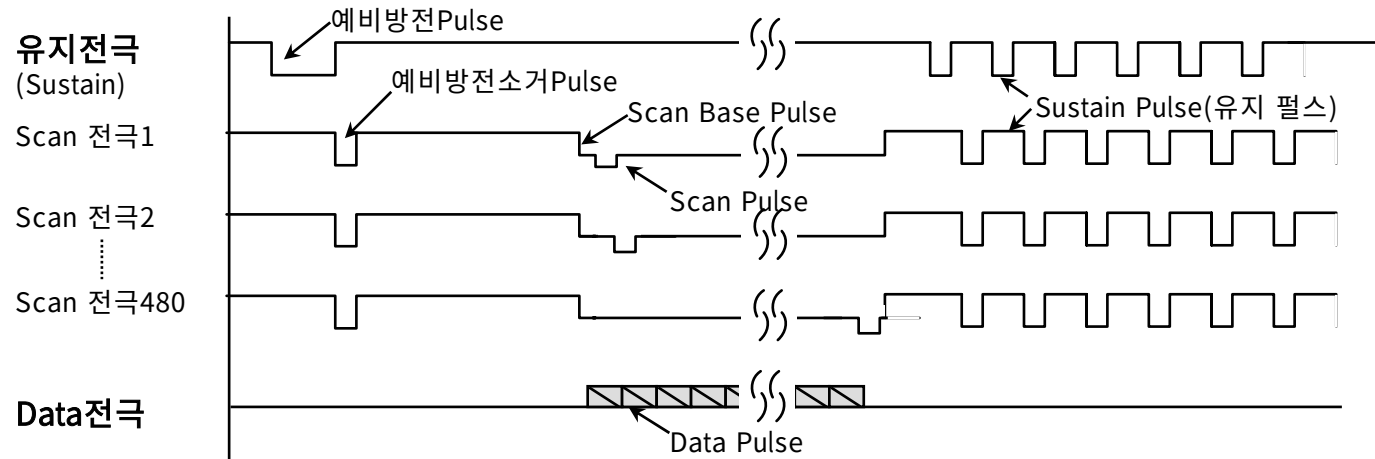


4. PDP Driver

AC 면 방전형 구동



4X4개의 방전 셀을 가진 PDP (상판 위에서 본 모양)



PDP의 구동 전압 파형

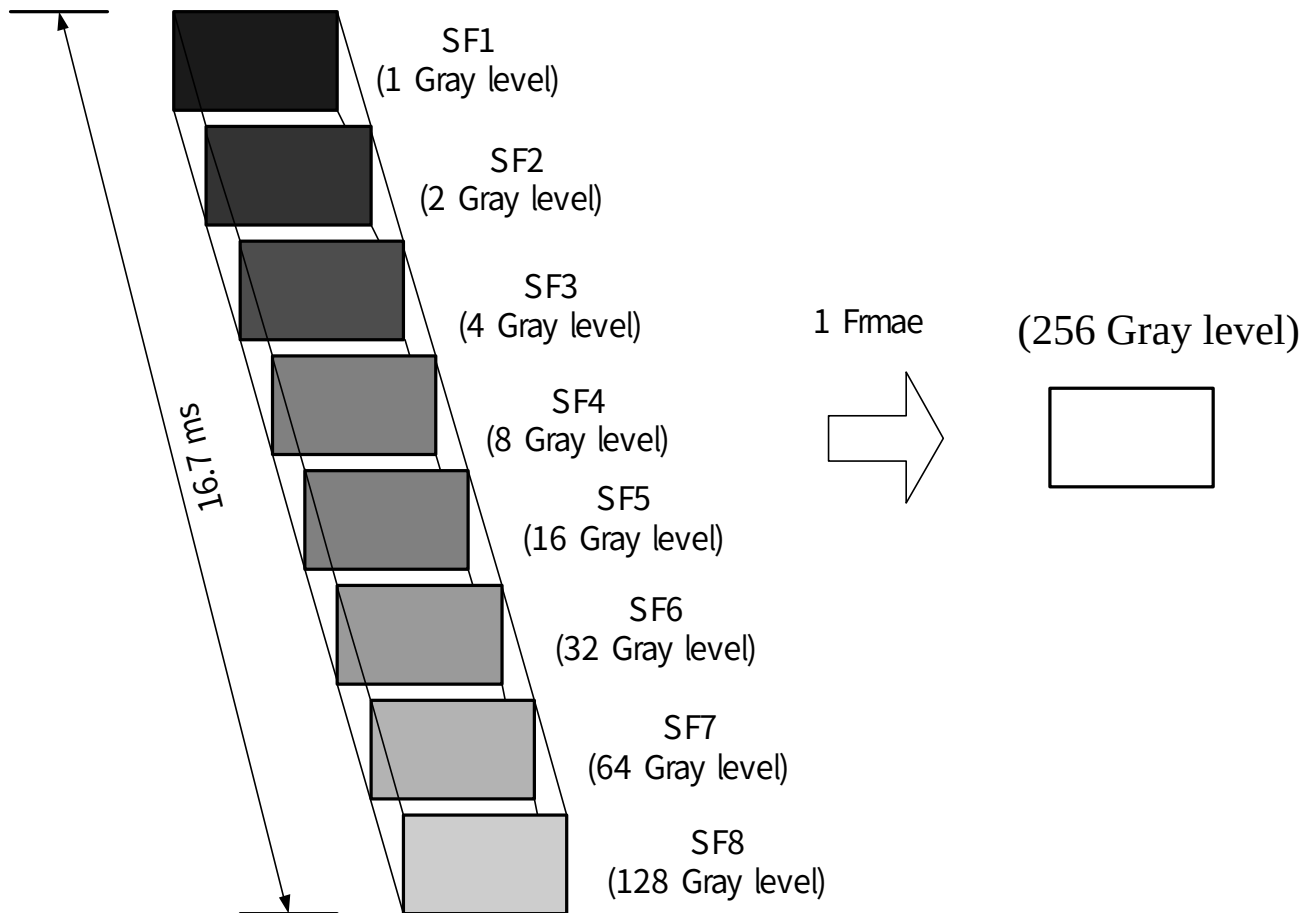
4. PDP Driver

PDP Data Driver의 구동 방식

- PDM (Pulse – Density Modulation) 방식 – 계조 표현 방식
 - Constant Voltage 구동
 - Variation of # of Pulse
 - > Gray역전현상
- ADS(Address Display Seperated)
 - Sub-Field Mapping
 - DA(Data Arranger)& TC(Timing Control)기능이 커짐

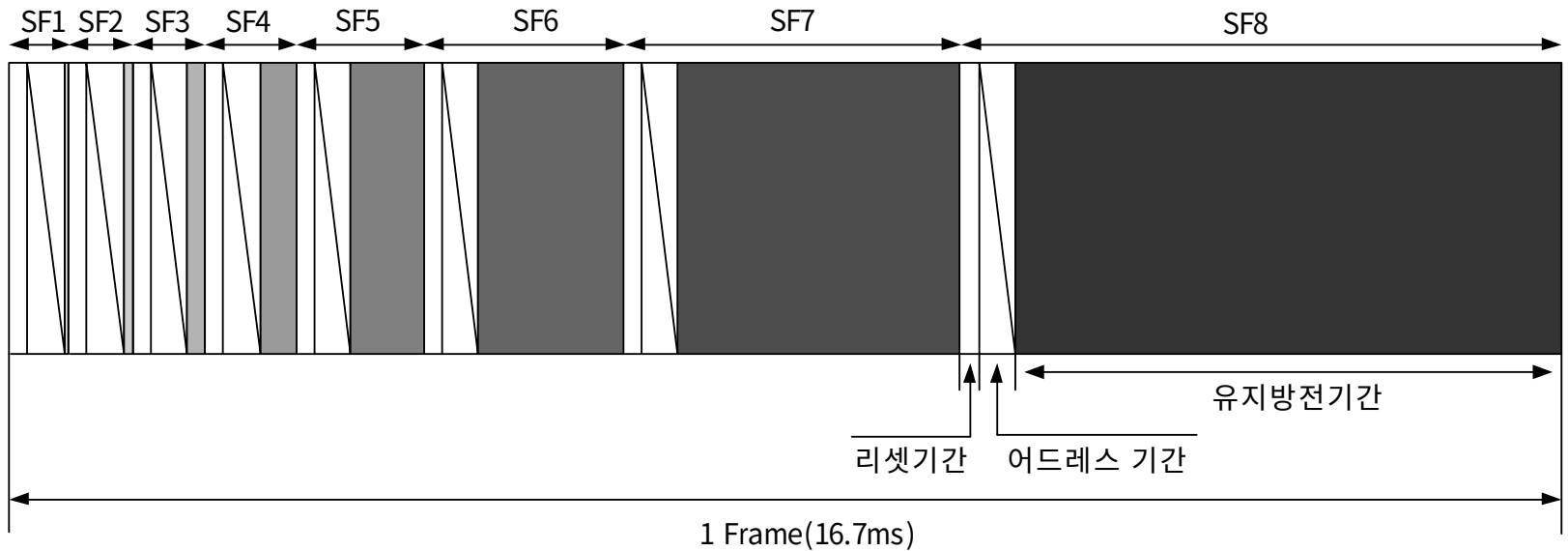
4. PDP Driver

PDP Data Driver의 구동 방식(256계조 구현)

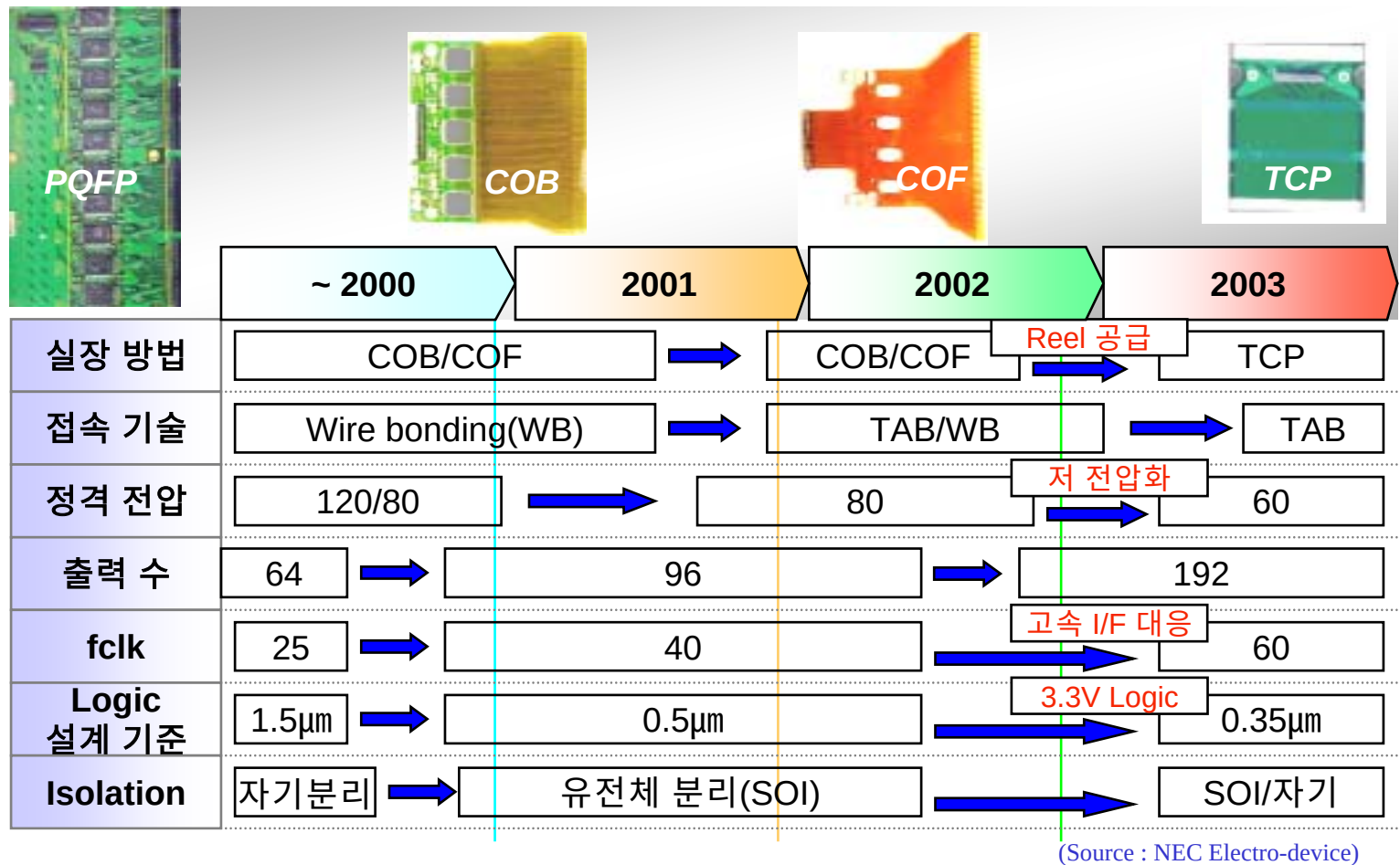


4. PDP Driver

PDP Data Driver의 구동 방식(계속)



4. PDP Driver (Data Driver IC 의 기술 Roadmap)

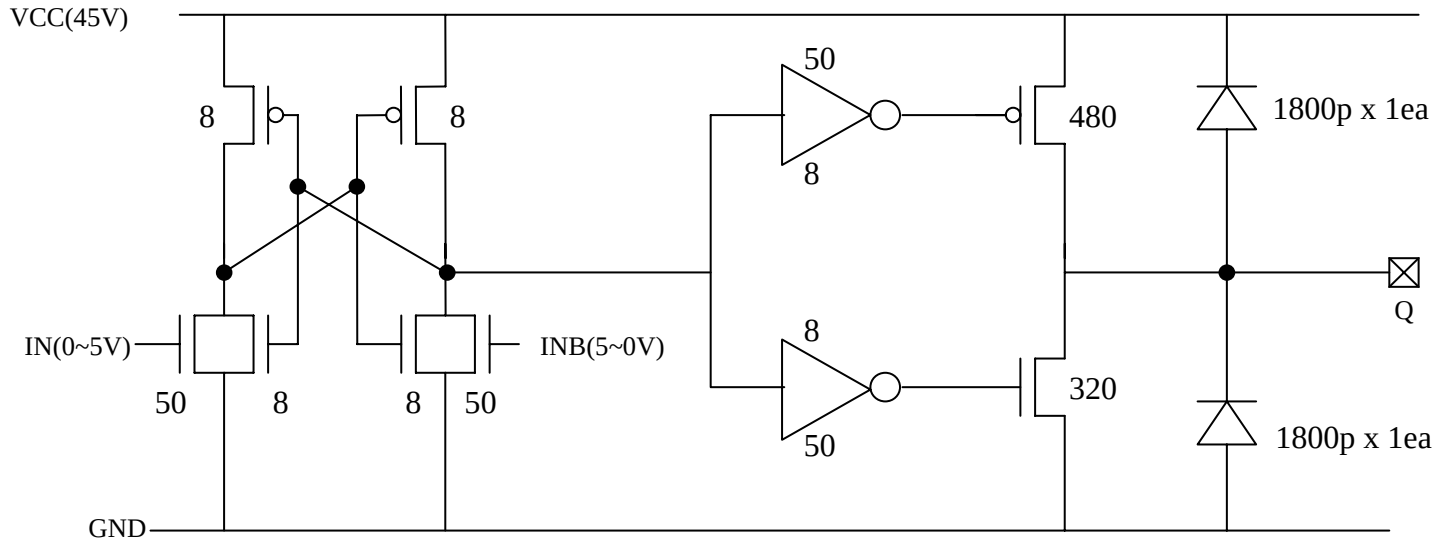


COB : Chip On Board, COM : Chip On Metal, COF : Chip On Film
TCP : Tape Carrier Package, TAB : Tape Automated Bonding, SOI : Silicon On Insulator

4. PDP Driver

OUTPUT Circuit Example 1.(50V공정)

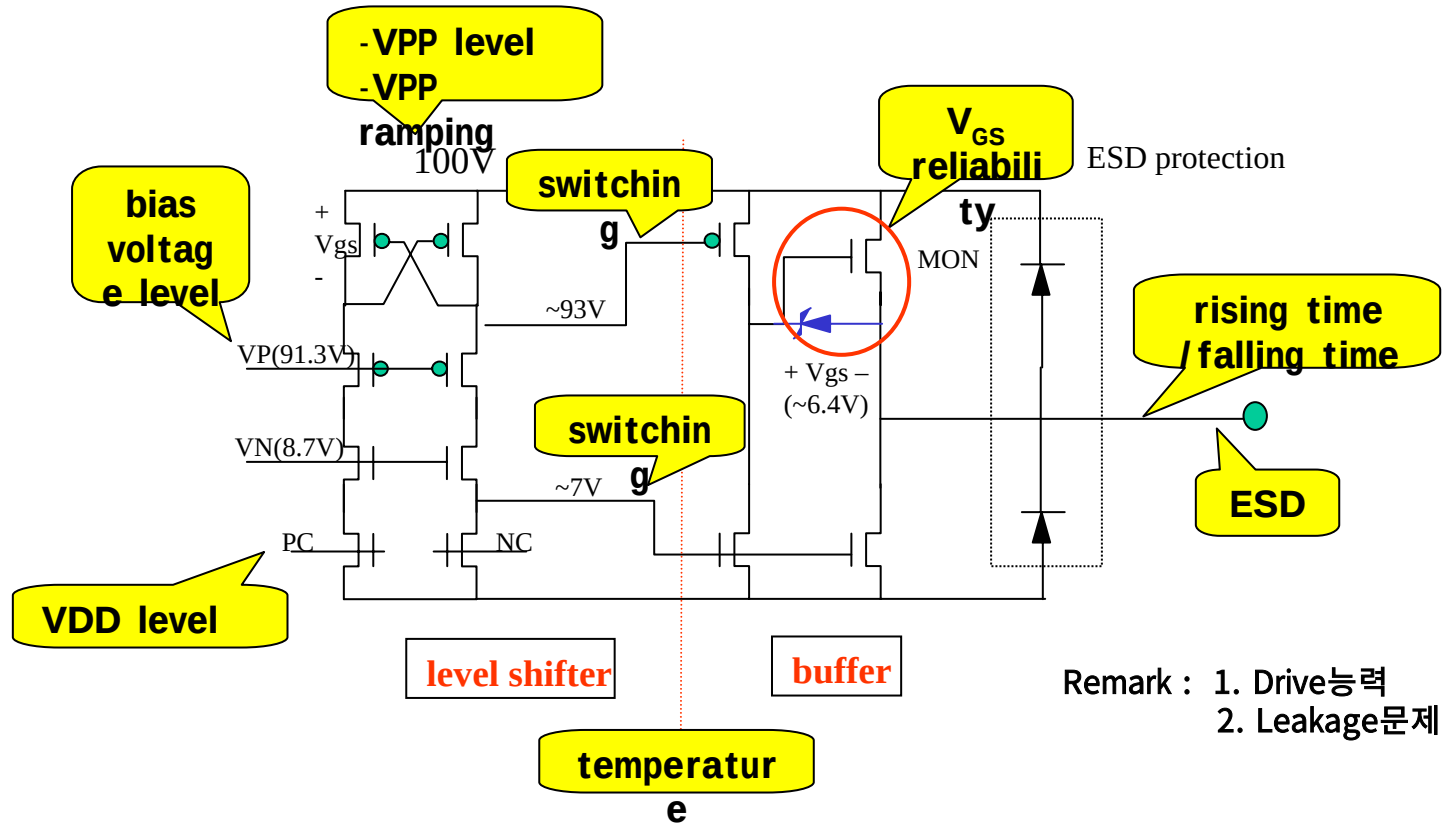
All Channel Length = 2um



- Remark :
1. Cross - Over(short) current
 2. Abnormal state current
 3. Output wave form

4. PDP Driver

OUTPUT Circuit Example 2.(100V공정 SOI)



4. PDP Driver

Data Driver IC의 기술 과제

- ✓ High Drivability
- ✓ Low Power Consumption(96Ch. -> 192Ch.)
- ✓ 고속 Interface 기술 (mini-LVDS, RSDS)
- ✓ Sufficient Operating Margin(Process)
- ✓ HV LDMOS / EDMOS / IGBT / SOI 공정의 이해
- ✓ HV Process Device의 ESD

Scan Driver IC의 기술 과제

- ✓ High Drivability
- ✓ Low Power Consumption(64Ch. -> 96Ch.)
- ✓ Sufficient Operating Margin(Process)

5. Conclusion

